

Muneeb Rehman
CT-25092

Lab 5

```
E:\University\University\SEMESTER-2\OOP\LAB\LAB-5\CT-25092_MuneebRehman - OOP_LAB_Assignment5.1.cpp - Embarcadero Dev-C++ 6.3
File Edit Search View Project Execute Tools AStyle Window Help
(globals)
Project Classes < > CT-25092_MuneebRehman - OOP_LAB_Assignment5.1.cpp CT-25092_MuneebRehman - OOP_LAB_Assignment5.2.cpp CT-25092_MuneebRehman - OOP_LAB_Assignment5.3.cpp CT-25092_MuneebRehman - OOP_LAB_Assignment5.4.cpp

1 // Exercise 1: Demonstrating public, protected, and private inheritance
2
3 #include <iostream>
4
5 class Base
6 {
7     private:
8         int privateInt;
9
10    protected:
11        int protectedInt;
12
13    public:
14        int publicInt;
15
16        int getPrivateInt()
17        {
18            return privateInt;
19        }
20
21        int getProtectedInt()
22        {
23            return protectedInt;
24        }
25
26        int getPublicInt()
27        {
28            return publicInt;
29        }
30
31        void setPrivateInt(int val)
32        {
33            privateInt = val;
34        }
35
36        void setProtectedInt(int val)
37        {
38            protectedInt = val;
39        }
40
41        void setPublicInt(int val)
42        {
43            publicInt = val;
44        }
45 };
46
47 class publicChild : public Base
48 {
49     public:
50        void displayAccess()
51        {
52            std::cout << "\n--- Public Inheritance ---";
53
54            std::cout << "\nPrivate Int (via getter): " << getPrivateInt();
55
56            std::cout << "\nProtected Int (direct access): " << protectedInt;
57
58            std::cout << "\nPublic Int (direct access): " << publicInt;
59        }
60 };
61
62 class protectedChild : protected Base
63 {
64     // ...
65 }
```

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Project Classes < > CT-25092_MuneebRehman - OOP_LAB_Assignment5.1.cpp CT-25092_MuneebRehman - OOP_LAB_Assignment5.2.cpp CT-25092_MuneebRehman - OOP_LAB_Assignment5.3.cpp CT-25092_MuneebRehman - OOP_LAB_Assignment5.4.cpp

64     public:
65         void displayAccess()
66         {
67             std::cout << "\n--- Protected Inheritance ---";
68
69             std::cout << "\nPrivate Int (via getter): " << getPrivateInt();
70
71             std::cout << "\nProtected Int (direct access): " << protectedInt;
72
73             std::cout << "\nPublic Int (direct access): " << publicInt;
74         }
75
76         void setValues(int priv, int prot, int pub)
77         {
78             setPrivateInt(priv);
79             setProtectedInt(prot);
80             setPublicInt(pub);
81         }
82     };
83
84 class privateChild : private Base
85 {
86     public:
87         void displayAccess()
88         {
89             std::cout << "\n--- Private Inheritance ---";
90
91             std::cout << "\nPrivate Int (via getter): " << getPrivateInt();
92
93             std::cout << "\nProtected Int (direct access): " << protectedInt;
94
95             std::cout << "\nPublic Int (direct access): " << publicInt;
96         }
97
98         void setValues(int priv, int prot, int pub)
99         {
100             setPrivateInt(priv);
101             setProtectedInt(prot);
102             setPublicInt(pub);
103         }
104     };
105
106 int main()
107 {
108     publicChild obj1;
109     obj1.setPrivateInt(10);
110     obj1.setProtectedInt(20);
111     obj1.setPublicInt(30);
112     obj1.displayAccess();
113
114     protectedChild obj2;
115     obj2.setValues(40, 50, 60);
116     obj2.displayAccess();
117
118     privateChild obj3;
119     obj3.setValues(70, 80, 90);
120     obj3.displayAccess();
121
122     std::cout << "\n\nSummary:";
123     std::cout << "\n\n- Public Inheritance: Public and Protected members accessible, Private needs getter";
124     std::cout << "\n\n- Protected Inheritance: All base members become protected in child";
125     std::cout << "\n\n- Private Inheritance: All base members become private in child\n";
126 }
```

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```
--- Public Inheritance ---  
Private Int (via getter): 10  
Protected Int (direct access): 20  
Public Int (direct access): 30
```

```
--- Protected Inheritance ---  
Private Int (via getter): 40  
Protected Int (direct access): 50  
Public Int (direct access): 60
```

```
--- Private Inheritance ---  
Private Int (via getter): 70  
Protected Int (direct access): 80  
Public Int (direct access): 90
```

Summary:

- Public Inheritance: Public and Protected members accessible, Private needs getter
- Protected Inheritance: All base members become protected in child
- Private Inheritance: All base members become private in child

```
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Process exited after 0.1742 seconds with return value 0  
Press any key to continue . . . |
```

```
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File Edit Search View Project Execute Tools AStyle Window Help
[Icons] [globals] TDM-GCC 9.2.0 64-bit Debug
Project Classes > [CT-25092_MuneebRehman - OOP_LAB_Assignment5.1.cpp] CT-25092_MuneebRehman - OOP_LAB_Assignment5.2.cpp CT-25092_MuneebRehman - OOP_LAB_Assignment5.3.cpp CT-25092_MuneebRehman - OOP_LAB_Assignment5.4.cpp

1 // Exercise 2: Teacher class with derived classes for different subjects
2
3 #include <iostream>
4 #include <string.h>
5
6 class Teacher
7 {
8 private:
9     char* Name;
10    int Age;
11    char* Institute;
12
13 public:
14    Teacher()
15    {
16        Name = NULL;
17        Age = 0;
18        Institute = NULL;
19    }
20
21    ~Teacher()
22    {
23        if (Name != NULL)
24        {
25            delete[] Name;
26        }
27        if (Institute != NULL)
28        {
29            delete[] Institute;
30        }
31    }
32
33    const char* getName()
34    {
35        return Name;
36    }
37
38    int getAge()
39    {
40        return Age;
41    }
42
43    const char* getInstitute()
44    {
45        return Institute;
46    }
47
48    void setName(const char* name)
49    {
50        if (Name != NULL)
51        {
52            delete[] Name;
53        }
54        Name = new char[strlen(name) + 1];
55        strcpy(Name, name);
56    }
57
58    void setAge(int age)
59    {
60        Age = age;
61    }
62
63    void setInstitute(const char* inst)
```

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File Edit Search View Project Execute Tools AStyle Window Help
[Icons] [globals] TDM-GCC 9.2.0 64-bit Debug
Project Classes > [CT-25092_MuneebRehman - OOP_LAB_Assignment5.1.cpp] CT-25092_MuneebRehman - OOP_LAB_Assignment5.2.cpp CT-25092_MuneebRehman - OOP_LAB_Assignment5.3.cpp CT-25092_MuneebRehman - OOP_LAB_Assignment5.4.cpp

61
62
63    void setInstitute(const char* inst)
64    {
65        if (Institute != NULL)
66        {
67            delete[] Institute;
68        }
69        Institute = new char[strlen(inst) + 1];
70        strcpy(Institute, inst);
71    }
72
73
74 class HumanitiesTeacher : public Teacher
75 {
76 private:
77     char Department[20];
78     char* CourseName;
79     char* Designation;
80
81 public:
82    HumanitiesTeacher()
83    {
84        strcpy(Department, "Humanities");
85        CourseName = NULL;
86        Designation = NULL;
87    }
88
89    ~HumanitiesTeacher()
90    {
91        if (CourseName != NULL)
92        {
93            delete[] CourseName;
94        }
95        if (Designation != NULL)
96        {
97            delete[] Designation;
98        }
99    }
100
101    const char* getDepartment()
102    {
103        return Department;
104    }
105
106    const char* getCourseName()
107    {
108        return CourseName;
109    }
110
111    const char* getDesignation()
112    {
113        return Designation;
114    }
115
116    void setCourseName(const char* course)
117    {
118        if (CourseName != NULL)
119        {
120            delete[] CourseName;
121        }
122        CourseName = new char[strlen(course) + 1];
123        strcpy(CourseName, course);
124    }
125}
```

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File Edit Search View Project Execute Tools AStyle Window Help
[Icons] [globals] [Debugger: TDM-GCC 9.2.0 64-bit Debug]
Project Classes > [CT-25092_MuneebRehman - OOP_LAB_Assignment5.1.cpp] [CT-25092_MuneebRehman - OOP_LAB_Assignment5.2.cpp] [CT-25092_MuneebRehman - OOP_LAB_Assignment5.3.cpp] [CT-25092_MuneebRehman - OOP_LAB_Assignment5.4.cpp]

121     }
122     CourseName = new char[strlen(course) + 1];
123     strcpy(CourseName, course);
124 }
125
126 void setDesignation(const char* desig)
127 {
128     if (Designation != NULL)
129     {
130         delete[] Designation;
131     }
132     Designation = new char[strlen(desig) + 1];
133     strcpy(Designation, desig);
134 }
135
136 void display()
137 {
138     std::cout << "\n--- Humanities Teacher Details ---";
139     std::cout << "\nName: " << getName();
140     std::cout << "\nAge: " << getAge();
141     std::cout << "\nInstitute: " << getInstitute();
142     std::cout << "\nDepartment: " << Department;
143     std::cout << "\nCourse Name: " << CourseName;
144     std::cout << "\nDesignation: " << Designation;
145     std::cout << "\n";
146 }
147
148 };
149
150 class ScienceTeacher : public Teacher
151 {
152 private:
153     char Department[20];
154     char* CourseName;
155     char* Designation;
156
157 public:
158     ScienceTeacher()
159     {
160         strcpy(Department, "Science");
161         CourseName = NULL;
162         Designation = NULL;
163     }
164     ~ScienceTeacher()
165     {
166         if (CourseName != NULL)
167         {
168             delete[] CourseName;
169         }
170         if (Designation != NULL)
171         {
172             delete[] Designation;
173         }
174     }
175     const char* getDepartment()
176     {
177         return Department;
178     }
179
180     const char* getCourseName()
181     {
182         return CourseName;
183     }
184 }
```

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[Icons] [globals] [Debugger: TDM-GCC 9.2.0 64-bit Debug]
Project Classes > [CT-25092_MuneebRehman - OOP_LAB_Assignment5.1.cpp] [CT-25092_MuneebRehman - OOP_LAB_Assignment5.2.cpp] [CT-25092_MuneebRehman - OOP_LAB_Assignment5.3.cpp] [CT-25092_MuneebRehman - OOP_LAB_Assignment5.4.cpp]

181     const char* getCourseName()
182     {
183         return CourseName;
184     }
185
186     const char* getDesignation()
187     {
188         return Designation;
189     }
190
191 void setCourseName(const char* course)
192 {
193     if (CourseName != NULL)
194     {
195         delete[] CourseName;
196     }
197     CourseName = new char[strlen(course) + 1];
198     strcpy(CourseName, course);
199 }
200
201 void setDesignation(const char* desig)
202 {
203     if (Designation != NULL)
204     {
205         delete[] Designation;
206     }
207     Designation = new char[strlen(desig) + 1];
208     strcpy(Designation, desig);
209 }
210
211 void display()
212 {
213     std::cout << "\n--- Science Teacher Details ---";
214     std::cout << "\nName: " << getName();
215     std::cout << "\nAge: " << getAge();
216     std::cout << "\nInstitute: " << getInstitute();
217     std::cout << "\nDepartment: " << Department;
218     std::cout << "\nCourse Name: " << CourseName;
219     std::cout << "\nDesignation: " << Designation;
220     std::cout << "\n";
221 }
222
223 };
224
225 class MathsTeacher : public Teacher
226 {
227 private:
228     char Department[20];
229     char* CourseName;
230     char* Designation;
231
232 public:
233     MathsTeacher()
234     {
235         strcpy(Department, "Maths");
236         CourseName = NULL;
237         Designation = NULL;
238     }
239     ~MathsTeacher()
240     {
241         if (CourseName != NULL)
242         {
243             delete[] CourseName;
244         }
245         if (Designation != NULL)
246         {
247             delete[] Designation;
248         }
249     }
250 }
```

```
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File Edit Search View Project Execute Tools AStyle Window Help
TDM-GCC 9.2.0 64-bit Debug
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Project Classes < > [CT-25092_MuneebRehman - OOP_LAB_Assignment5.1.cpp] CT-25092_MuneebRehman - OOP_LAB_Assignment5.2.cpp CT-25092_MuneebRehman - OOP_LAB_Assignment5.3.cpp CT-25092_MuneebRehman - OOP_LAB_Assignment5.4.cpp

241 {
242     if (CourseName != NULL)
243     {
244         delete[] CourseName;
245     }
246     if (Designation != NULL)
247     {
248         delete[] Designation;
249     }
250 }
251
252 const char* getDepartment()
253 {
254     return Department;
255 }
256
257 const char* getCourseName()
258 {
259     return CourseName;
260 }
261
262 const char* getDesignation()
263 {
264     return Designation;
265 }
266
267 void setCourseName(const char* course)
268 {
269     if (CourseName != NULL)
270     {
271         delete[] CourseName;
272     }
273     CourseName = new char[strlen(course) + 1];
274     strcpy(CourseName, course);
275 }
276
277 void setDesignation(const char* desig)
278 {
279     if (Designation != NULL)
280     {
281         delete[] Designation;
282     }
283     Designation = new char[strlen(desig) + 1];
284     strcpy(Designation, desig);
285 }
286
287 void display()
288 {
289     std::cout << "\n--- Maths Teacher Details ---";
290     std::cout << "\nName: " << getName();
291     std::cout << "\nAge: " << getAge();
292     std::cout << "\nInstitute: " << getInstitute();
293     std::cout << "\nDepartment: " << Department;
294     std::cout << "\nCourse Name: " << CourseName;
295     std::cout << "\nDesignation: " << Designation;
296     std::cout << "\n";
297 }
298
299 int main()
300 {
301     char tempInput[100];
302     int ageInput;
303 }
```

```
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301 char tempInput[100];
302 int ageInput;
303
304 HumanitiesTeacher teacher1;
305 std::cout << "\n--- Enter Humanities Teacher Details ---\n";
306 std::cout << "Name: ";
307 std::cin.getline(tempInput, 100);
308 teacher1.setName(tempInput);
309
310 std::cout << "Age: ";
311 std::cin >> ageInput;
312 teacher1.setAge(ageInput);
313 std::cin.ignore();
314
315 std::cout << "Institute: ";
316 std::cin.getline(tempInput, 100);
317 teacher1.setInstitute(tempInput);
318
319 std::cout << "Course Name: ";
320 std::cin.getline(tempInput, 100);
321 teacher1.setCourseName(tempInput);
322
323 std::cout << "Designation: ";
324 std::cin.getline(tempInput, 100);
325 teacher1.setDesignation(tempInput);
326
327 ScienceTeacher teacher2;
328 std::cout << "\n--- Enter Science Teacher Details ---\n";
329 std::cout << "Name: ";
330 std::cin.getline(tempInput, 100);
331 teacher2.setName(tempInput);
332
333 std::cout << "Age: ";
334 std::cin >> ageInput;
335 teacher2.setAge(ageInput);
336 std::cin.ignore();
337
338 std::cout << "Institute: ";
339 std::cin.getline(tempInput, 100);
340 teacher2.setInstitute(tempInput);
341
342 std::cout << "Course Name: ";
343 std::cin.getline(tempInput, 100);
344 teacher2.setCourseName(tempInput);
345
346 std::cout << "Designation: ";
347 std::cin.getline(tempInput, 100);
348 teacher2.setDesignation(tempInput);
349
350 MathsTeacher teacher3;
351 std::cout << "\n--- Enter Maths Teacher Details ---\n";
352 std::cout << "Name: ";
353 std::cin.getline(tempInput, 100);
354 teacher3.setName(tempInput);
355
356 std::cout << "Age: ";
357 std::cin >> ageInput;
358 teacher3.setAge(ageInput);
359 std::cin.ignore();
360
361 std::cout << "Institute: ";
362 std::cin.getline(tempInput, 100);
363 teacher3.setInstitute(tempInput);
364
365 std::cout << "Course Name: ";
366 std::cin.getline(tempInput, 100);
367 teacher3.setCourseName(tempInput);
368
369 std::cout << "Designation: ";
370 std::cin.getline(tempInput, 100);
371 teacher3.setDesignation(tempInput);
372 }
```

The image shows a C++ IDE with a project named "CT-25092_MuneebRehman - OOP_LAB_Assignment5.2.cpp". The code is written in C++ and uses the `std::cin` and `std::cout` streams for input and output. It defines three classes: `Teacher`, `ScienceTeacher`, and `MathsTeacher`. The `Teacher` class has attributes `name`, `age`, and `institute`. The `ScienceTeacher` and `MathsTeacher` classes inherit from `Teacher` and have an additional attribute `courseName`. The program prompts the user to enter details for three teachers and then displays the information for each.

```
1  E:\University\SEMESTER-2\OOP\LAB\LAB-5\CT-25092_MuneebRehman - OOP_LAB_Assignment5.2.cpp - Embarcadero Dev-C++ 6.3
2  File Edit Search View Project Execute Tools ASyntax Window Help
3  [Icons] | TDM-GCC 9.2.0 64-bit Debug
4  [Icons] | (globals)
5  Project Classes < > | CT-25092_MuneebRehman - OOP_LAB_Assignment5.1.cpp | CT-25092_MuneebRehman - OOP_LAB_Assignment5.2.cpp | CT-25092_MuneebRehman - OOP_LAB_Assignment5.3.cpp | CT-25092_MuneebRehman - OOP_LAB_Assignment5.4.cpp
6
7  318 teacher1.setInstitute(tempInput);
8  319
9  320 std::cout << "Course Name: ";
10 321 std::cin.getline(tempInput, 100);
11 322 teacher1.setCourseName(tempInput);
12 323
13 324 std::cout << "Designation: ";
14 325 std::cin.getline(tempInput, 100);
15 326 teacher1.setDesignation(tempInput);
16 327
17 328 ScienceTeacher teacher2;
18 329 std::cout << "\n=== Enter Science Teacher Details ===\n";
19 330 std::cout << "Name: ";
20 331 std::cin.getline(tempInput, 100);
21 332 teacher2.setName(tempInput);
22 333
23 334 std::cout << "Age: ";
24 335 std::cin >> ageInput;
25 336 teacher2.setAge(ageInput);
26 337 std::cin.ignore();
27 338
28 339 std::cout << "Institute: ";
29 340 std::cin.getline(tempInput, 100);
30 341 teacher2.setInstitute(tempInput);
31 342
32 343 std::cout << "Course Name: ";
33 344 std::cin.getline(tempInput, 100);
34 345 teacher2.setCourseName(tempInput);
35 346
36 347 std::cout << "Designation: ";
37 348 std::cin.getline(tempInput, 100);
38 349 teacher2.setDesignation(tempInput);
39 350
40 351 MathsTeacher teacher3;
41 352 std::cout << "\n=== Enter Maths Teacher Details ===\n";
42 353 std::cout << "Name: ";
43 354 std::cin.getline(tempInput, 100);
44 355 teacher3.setName(tempInput);
45 356
46 357 std::cout << "Age: ";
47 358 std::cin >> ageInput;
48 359 teacher3.setAge(ageInput);
49 360 std::cin.ignore();
50 361
51 362 std::cout << "Institute: ";
52 363 std::cin.getline(tempInput, 100);
53 364 teacher3.setInstitute(tempInput);
54 365
55 366 std::cout << "Course Name: ";
56 367 std::cin.getline(tempInput, 100);
57 368 teacher3.setCourseName(tempInput);
58 369
59 370 std::cout << "Designation: ";
60 371 std::cin.getline(tempInput, 100);
61 372 teacher3.setDesignation(tempInput);
62 373
63 374 std::cout << "\n=== Teacher Information ===";
64 375 teacher1.display();
65 376 teacher2.display();
66 377 teacher3.display();
67 378
68 379 return 0;
69 380
```

```

E:\University\University\SEME  X  +  ~
=== Enter Humanities Teacher Details ===
Name: John Doe
Age: 32
Institute: NED University
Course Name: Ethical Behaviour
Designation: Assistant Professor

=== Enter Science Teacher Details ===
Name: John Smith
Age: 29
Institute: NED University
Course Name: Physics
Designation: Lecturer

=== Enter Maths Teacher Details ===
Name: Jane Doe
Age: 30
Institute: NED University
Course Name: Mathematics
Designation: Lecturer

=== Teacher Information ===
--- Humanities Teacher Details ---
Name: John Doe
Age: 32
Institute: NED University
Department: Humanities
Course Name: Ethical Behaviour
Designation: Assistant Professor

--- Science Teacher Details ---
Name: John Smith
Age: 29
Institute: NED University
Department: Science
Course Name: Physics
Designation: Lecturer

--- Maths Teacher Details ---
Name: Jane Doe
Age: 30
Institute: NED University
Department: Maths
Course Name: Mathematics
Designation: Lecturer

-----
Process exited after 142.6 seconds with return value 0
Press any key to continue . . .

```

```
E:\University\University\SEMESTER-2\OOP\LAB\LAB-5\CT-25092_MuneebRehman - OOP_LAB_Assignment5.3.cpp - Embarcadero Dev-C++ 6.3
File Edit Search View Project Execute Tools AStyle Window Help
(globals)
Project Classes <> [CT-25092_MuneebRehman - OOP_LAB_Assignment5.1.cpp CT-25092_MuneebRehman - OOP_LAB_Assignment5.2.cpp CT-25092_MuneebRehman - OOP_LAB_Assignment5.3.cpp CT-25092_MuneebRehman - OOP_LAB_Assignment5.4.cpp]
1 // Exercise 3: Weapons Hierarchy - Multilevel Inheritance
2 #include <iostream>
3
4 class Weapons
5 {
6 public:
7     void WeaponsDescription()
8     {
9         std::cout << "Weapons: Used for combat and defense purposes.";
10    }
11 };
12
13 class HotWeapons : public Weapons
14 {
15 public:
16     void HotWeaponsDescription()
17     {
18         std::cout << "Hot Weapons: Uses gunpowder or explodes to cause damage.";
19    }
20 };
21
22 class Bombs : public HotWeapons
23 {
24 public:
25     void BombsDescription()
26     {
27         std::cout << "Bombs: Explosive devices that blow up on detonation.";
28    }
29 };
30
31 class NuclearBombs : public Bombs
32 {
33 public:
34     void NuclearBombsDescription()
35     {
36         std::cout << "Nuclear Bombs: Extremely powerful bombs that blow up using nuclear fission and fusion reactions.";
37    }
38 };
39
40 int main()
41 {
42     std::cout << "=== Defense Organization Weapons Hierarchy ===\n";
43
44     Weapons weapon1;
45     std::cout << "---- Level 1: Weapons ----\n";
46     weapon1.WeaponsDescription();
47
48     HotWeapons weapon2;
49     std::cout << "---- Level 2: Hot Weapons ----\n";
50     weapon2.WeaponsDescription();
51     weapon2.HotWeaponsDescription();
52
53     Bombs weapon3;
54     std::cout << "---- Level 3: Bombs ----\n";
55     weapon3.WeaponsDescription();
56     weapon3.HotWeaponsDescription();
57     weapon3.BombsDescription();
58
59     NuclearBombs weapon4;
60     std::cout << "---- Level 4: Nuclear Bombs ----\n";
61     weapon4.WeaponsDescription();
62     weapon4.HotWeaponsDescription();
63     weapon4.BombsDescription();
64     weapon4.NuclearBombsDescription();
65
66     std::cout << "Hierarchy: Weapons -> Hot Weapons -> Bombs -> Nuclear Bombs\n";
67 }
```

```
E:\University\University\SEME
=== Defense Organization Weapons Hierarchy ===

--- Level 1: Weapons ---
Weapons: Used for combat and defense purposes.

--- Level 2: Hot Weapons ---
Weapons: Used for combat and defense purposes.
Hot Weapons: Uses gunpowder or explodes to cause damage.

--- Level 3: Bombs ---
Weapons: Used for combat and defense purposes.
Hot Weapons: Uses gunpowder or explodes to cause damage.
Bombs: Explosive devices that blow up on detonation.

--- Level 4: Nuclear Bombs ---
Weapons: Used for combat and defense purposes.
Hot Weapons: Uses gunpowder or explodes to cause damage.
Bombs: Explosive devices that blow up on detonation.
Nuclear Bombs: Extremely powerful bombs that blow up using nuclear fission and fusion reactions.

Hierarchy: Weapons -> Hot Weapons -> Bombs -> Nuclear Bombs

-----

Process exited after 0.02578 seconds with return value 0
Press any key to continue . . . |
```


The image shows a C++ IDE interface. At the top, the file path is "E:\University\University\SEHSTER-2\OOP\LAB\LAB-5\CT-25092_MuneebRehman - OOP_LAB_Assignment5.4.cpp - Embarcadero Dev-C++ 6.3". The menu bar includes File, Edit, Search, View, Project, Execute, Tools, AStyle, Window, and Help. The toolbar contains icons for file operations and compilation. The project explorer on the left shows a project named "CT-25092_MuneebRehman" with files "CT-25092_MuneebRehman - OOP_LAB_Assignment5.1.cpp", "CT-25092_MuneebRehman - OOP_LAB_Assignment5.2.cpp", "CT-25092_MuneebRehman - OOP_LAB_Assignment5.3.cpp", and "CT-25092_MuneebRehman - OOP_LAB_Assignment5.4.cpp". The code editor displays the content of "CT-25092_MuneebRehman - OOP_LAB_Assignment5.1.cpp". The code is a C++ program for a bakery billing system with inheritance. It includes `<iostream>` and `<string.h>`. It defines a base class `Item` with a protected `char* Name` and `int Quantity`. The `Item` class has a constructor, a destructor, and methods `setName`, `setQuantity`, `getName`, `getQuantity`, and `calculateTotal`. A derived class `BakedGoods` inherits from `Item` and has a protected `float Discount`.

```
1 // Exercise 4: Bakery billing system with inheritance
2
3 #include <iostream>
4 #include <string.h>
5
6 class Item
7 {
8     protected:
9         char* Name;
10        int Quantity;
11
12    public:
13        Item()
14        {
15            Name = NULL;
16            Quantity = 0;
17        }
18
19        ~Item()
20        {
21            if (Name != NULL)
22            {
23                delete[] Name;
24            }
25        }
26
27        void setName(const char* name)
28        {
29            if (Name != NULL)
30            {
31                delete[] Name;
32            }
33            Name = new char[strlen(name) + 1];
34            strcpy(Name, name);
35        }
36
37        void setQuantity(int qty)
38        {
39            Quantity = qty;
40        }
41
42        const char* getName()
43        {
44            return Name;
45        }
46
47        int getQuantity()
48        {
49            return Quantity;
50        }
51
52        virtual float calculateTotal()
53        {
54            return 0.0;
55        }
56    };
57
58 class BakedGoods : public Item
59 {
60     protected:
61         float Discount;
62
63     public:
64         BakedGoods()
65         {
66             Discount = 0.0;
67         }
68
69         ~BakedGoods()
70         {
71             // No need to delete anything here
72         }
73
74         void setName(const char* name)
75         {
76             Item::setName(name);
77         }
78
79         void setQuantity(int qty)
80         {
81             Item::setQuantity(qty);
82         }
83
84         const char* getName()
85         {
86             return Item::getName();
87         }
88
89         int getQuantity()
90         {
91             return Item::getQuantity();
92         }
93
94         virtual float calculateTotal()
95         {
96             return Item::calculateTotal() + Discount;
97         }
98
99         void setDiscount(float discount)
100        {
101            Discount = discount;
102        }
103
104        float getDiscount()
105        {
106            return Discount;
107        }
108    };
109
110 int main()
111 {
112     Item* item = new Item();
113     item->setName("Bread");
114     item->setQuantity(10);
115     float total = item->calculateTotal();
116     cout << "Total: " << total << endl;
117
118     BakedGoods* bakedGoods = new BakedGoods();
119     bakedGoods->setName("Cake");
120     bakedGoods->setQuantity(5);
121     bakedGoods->setDiscount(10.0);
122     total = bakedGoods->calculateTotal();
123     cout << "Total: " << total << endl;
124
125     delete item;
126     delete bakedGoods;
127
128     return 0;
129 }
```

The screenshot shows a C++ IDE with a project named "Project" containing several files. The active file is "CT-25092_MuneebRehman - OOP_LAB_Assignment5.4.cpp". The code defines a base class "BakedGoods" and two subclasses, "Cakes" and "Bread".

```
61 float Discount;
62
63 public:
64     BakedGoods()
65     {
66         Discount = 0.10;
67     }
68
69     float getDiscount()
70     {
71         return Discount;
72     }
73 };
74
75 class Cakes : public BakedGoods
76 {
77 private:
78     float Price;
79
80 public:
81     Cakes()
82     {
83         Price = 600.0;
84         setName("Cake");
85     }
86
87     float getPrice()
88     {
89         return Price;
90     }
91
92     float calculateTotal()
93     {
94         float total = Price * Quantity;
95         float discountAmount = total * Discount;
96         return total - discountAmount;
97     }
98
99     void displayBill()
100     {
101         std::cout << "Item: " << Name;
102         std::cout << "\nQuantity: " << Quantity;
103         std::cout << "\nPrice per unit: " << Price;
104         std::cout << "\nDiscount: " << (Discount * 100) << "%";
105         std::cout << "\nTotal: " << calculateTotal() << " PKR";
106     }
107 };
108
109 class Bread : public BakedGoods
110 {
111 private:
112     float Price;
113
114 public:
115     Bread()
116     {
117         Price = 200.0;
118         setName("Bread");
119     }
120
121     float getPrice()
122     {
123         return Price;
```

```
E:\University\University\SEMESTER-2\OOP\LAB\LAB-5\CT-25092_MuneebRehman - OOP_LAB_Assignment5.4.cpp - Embarcadero Dev-C++ 6.3
File Edit Search View Project Execute Tools AStyle Window Help
[Icons] [globals] [Debugger: TDM-GCC 9.2.0 64-bit Debug]
Project Classes < > [CT-25092_MuneebRehman - OOP_LAB_Assignment5.1.cpp] CT-25092_MuneebRehman - OOP_LAB_Assignment5.2.cpp CT-25092_MuneebRehman - OOP_LAB_Assignment5.3.cpp CT-25092_MuneebRehman - OOP_LAB_Assignment5.4.cpp
121 float getPrice()
122 {
123     return Price;
124 }
125
126 float calculateTotal()
127 {
128     float total = Price * Quantity;
129     float discountAmount = total * Discount;
130     return total - discountAmount;
131 }
132
133 void displayBill()
134 {
135     std::cout << "Name: " << Name;
136     std::cout << "Quantity: " << Quantity;
137     std::cout << "Price per unit: " << Price;
138     std::cout << "Discount: " << (Discount * 100) << "%";
139     std::cout << "Total: " << calculateTotal() << " PKR";
140 }
141 };
142
143 class Drinks : public Item
144 {
145 private:
146     float Discount;
147     float Price;
148 public:
149     Drinks()
150     {
151         Discount = 0.05;
152         Price = 100.0;
153         setName("Drink");
154     }
155
156     float getDiscount()
157     {
158         return Discount;
159     }
160
161     float getPrice()
162     {
163         return Price;
164     }
165
166     float calculateTotal()
167     {
168         float total = Price * Quantity;
169         float discountAmount = total * Discount;
170         return total - discountAmount;
171     }
172
173     void displayBill()
174     {
175         std::cout << "Name: " << Name;
176         std::cout << "Quantity: " << Quantity;
177         std::cout << "Price per unit: " << Price;
178         std::cout << "Discount: " << (Discount * 100) << "%";
179         std::cout << "Total: " << calculateTotal() << " PKR";
180     }
181 };
182
183
```

```
E:\University\University\SEMESTER-2\OOP\LAB\LAB-5\CT-25092_MuneebRehman - OOP_LAB_Assignment5.4.cpp - Embarcadero Dev-C++ 6.3
File Edit Search View Project Execute Tools AStyle Window Help
[Icons] [globals] [Debugger: TDM-GCC 9.2.0 64-bit Debug]
Project Classes < > [CT-25092_MuneebRehman - OOP_LAB_Assignment5.1.cpp] CT-25092_MuneebRehman - OOP_LAB_Assignment5.2.cpp CT-25092_MuneebRehman - OOP_LAB_Assignment5.3.cpp CT-25092_MuneebRehman - OOP_LAB_Assignment5.4.cpp
168 {
169     float total = Price * Quantity;
170     float discountAmount = total * Discount;
171     return total - discountAmount;
172 }
173
174 void displayBill()
175 {
176     std::cout << "Name: " << Name;
177     std::cout << "Quantity: " << Quantity;
178     std::cout << "Price per unit: " << Price;
179     std::cout << "Discount: " << (Discount * 100) << "%";
180     std::cout << "Total: " << calculateTotal() << " PKR";
181 }
182 };
183
184 int main()
185 {
186     int numCakes, numBread, numDrinks;
187     float grandTotal = 0.0;
188
189     std::cout << "=== Bakery Billing System ===\n";
190
191     std::cout << "Enter quantity of Cakes: ";
192     std::cin >> numCakes;
193     Cakes cakeOrder;
194     cakeOrder.setQuantity(numCakes);
195
196     std::cout << "Enter quantity of Bread: ";
197     std::cin >> numBread;
198     Bread breadOrder;
199     breadOrder.setQuantity(numBread);
200
201     std::cout << "Enter quantity of Drinks: ";
202     std::cin >> numDrinks;
203     Drinks drinkOrder;
204     drinkOrder.setQuantity(numDrinks);
205
206     std::cout << "\n=== Customer Bill ===";
207
208     if (numCakes > 0)
209     {
210         std::cout << "\n";
211         cakeOrder.displayBill();
212         grandTotal = grandTotal + cakeOrder.calculateTotal();
213     }
214
215     if (numBread > 0)
216     {
217         std::cout << "\n";
218         breadOrder.displayBill();
219         grandTotal = grandTotal + breadOrder.calculateTotal();
220     }
221
222     if (numDrinks > 0)
223     {
224         std::cout << "\n";
225         drinkOrder.displayBill();
226         grandTotal = grandTotal + drinkOrder.calculateTotal();
227     }
228
229     std::cout << "\n\n--- Grand Total: " << grandTotal << " PKR ---\n";
230 }
```

```
E:\University\University\SEME x + v
=== Bakery Billing System ===

Enter quantity of Cakes: 24
Enter quantity of Bread: 32
Enter quantity of Drinks: 80

=== Customer Bill ===

Item: Cake
Quantity: 24
Price per unit: 600
Discount: 10%
Total: 12960 PKR

Item: Bread
Quantity: 32
Price per unit: 200
Discount: 10%
Total: 5760 PKR

Item: Drink
Quantity: 80
Price per unit: 100
Discount: 5%
Total: 7600 PKR

--- Grand Total: 26320 PKR ---

-----
Process exited after 17.74 seconds with return value 0
Press any key to continue . . . |
```